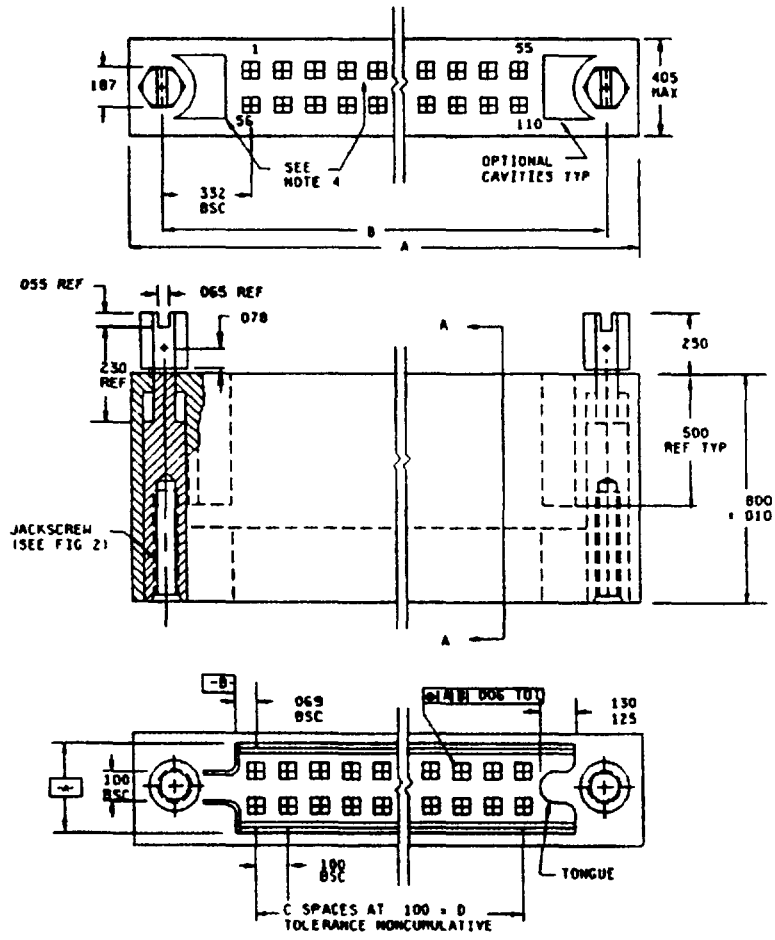
INCH-POUND
MIL-C-55302/113E(USAF)
6 MAY 1993
SUPERSEDING
MIL-C-55302/113D(USAF)
21 September 1988

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES. PLUG, STRAIGHT-THRU, 30 THROUGH 110 CONTACT POSITIONS, CRIMP REMOVABLE CONTACTS (.100 SQUARE GRID)

This specification sheet is approved for use by Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55302.

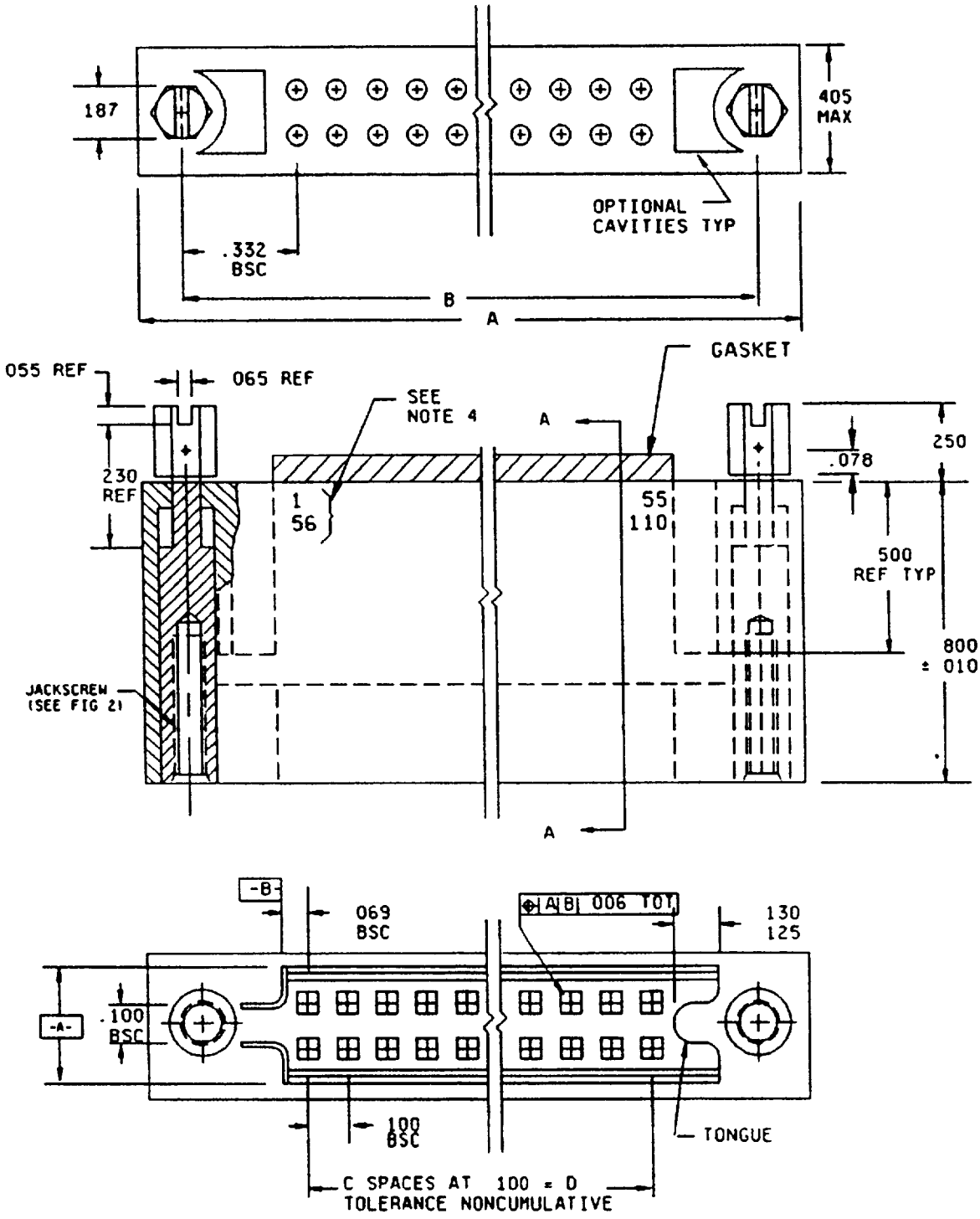


(E) FIGURE 1 Connectors, housing, plug, 100 (2.54 mm) square grid

(E) denotes changes

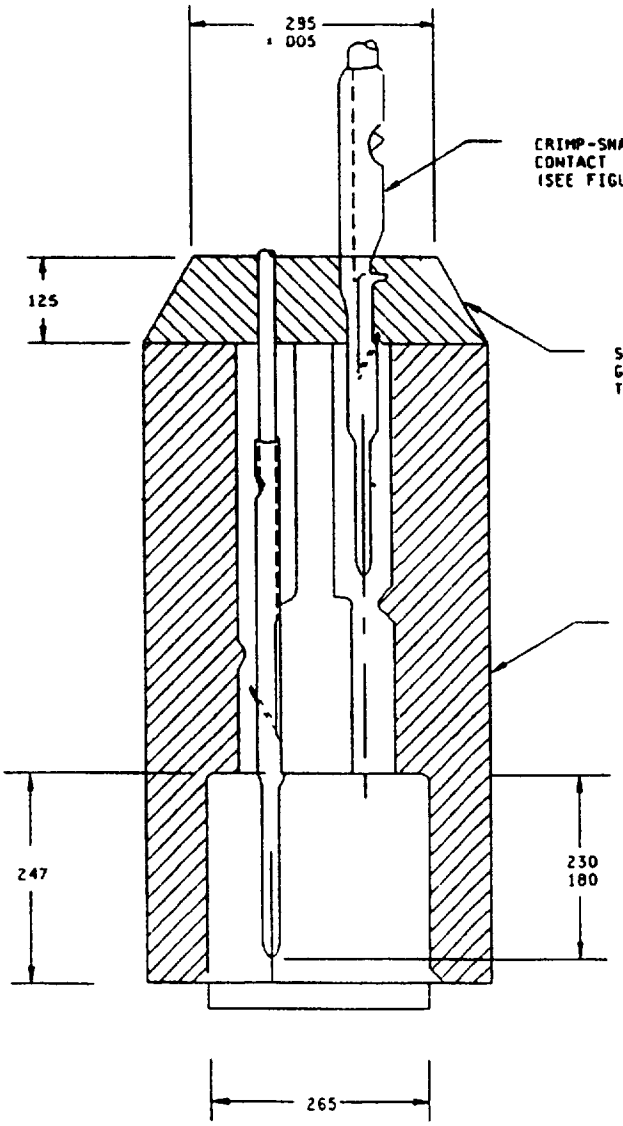
DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

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(E) FIGURE 1 Connectors, housing, plug, 100 (2.54 mm) square grid - Continued

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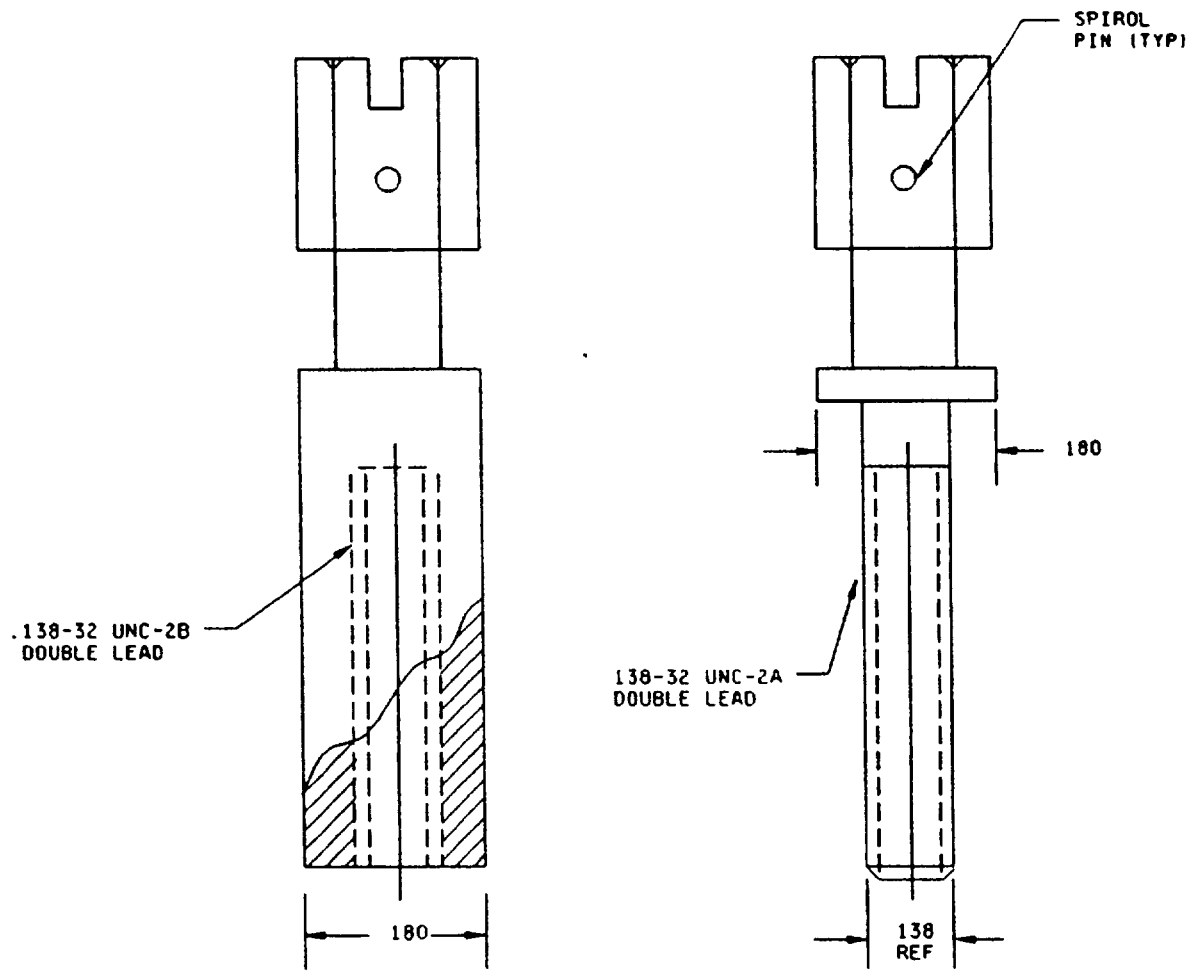
SECTION A-A

Inches	mm
005	13
006	15
010	25
015	38
055	1 40
065	1 65
069	1 75
.078	1.98
100	2.54
125	3 18
130	3 30
135	3 43
140	3 56
180	4 57
230	5 84
247	6 27
.250	6 35
265	6 73
332	8 43
400	10 16
800	20 32
3 400	86 36
4 065	103 25
4 355	110 62

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ± 005 (13 mm) on decimals
4. Numbers indicating end cavities and lines indicating every fifth cavity are stamped on housing

(E) FIGURE 1 Connector, housing, plug, 100 (2 54 mm) square grid - Continued

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FEMALE
M55302/113-11

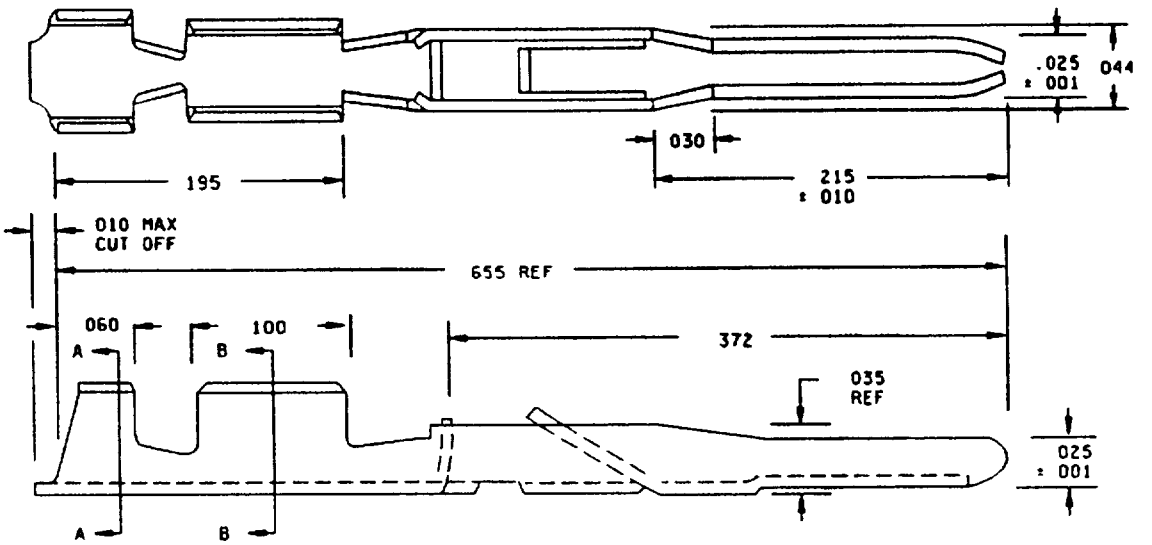
MALE
M55302/113-12

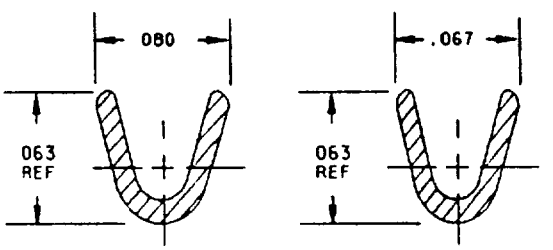
Inches	mm
138	3 51
180	4 57

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ± 005 (13 mm) on decimals

FIGURE 2 Turnable jackscrews

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SECTION A-A SECTION B-B

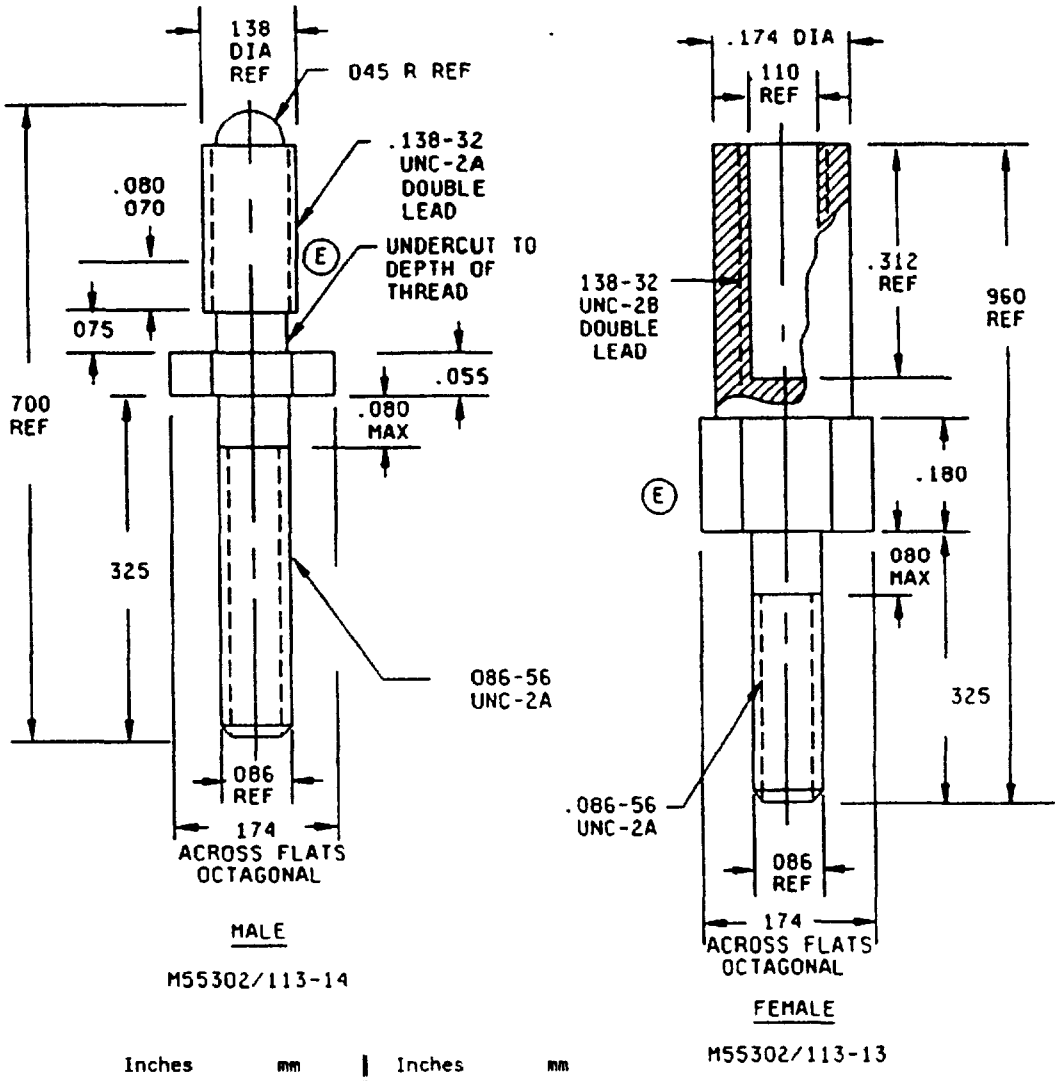
Inches	mm	Inches	mm
001	03	067	1 70
025	64	080	2 03
030	76	100	2 54
035	.89	195	4 95
044	1 12	215	5 46
060	1.52	372	9 45
.063	1 60	655	16 61

NOTES

1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) on decimals
4. Contact crimp for use with (AWG) wire range no 22 to no 26, insulation range 036 (.91 mm) to 054 (1 37 mm)
5. (E) Contact supplied loose piece, unless otherwise specified, part number M55302/113-05 Contacts may be supplied on reel form part number M55302/113-06

FIGURE 3 Pin contact, crimp-snap

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Inches	mm	Inches	mm
045	1 14	138	3 51
.055	1 40	180	4 57
080	2 03	.325	8 26
086	2 18	375	9 52
110	2 79	700	17 78
135	3 43	.960	24 38

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ± 005 (13 mm) on decimals.
4. This connector mates with the external key type connectors specified in MIL-C-55302/27 and is equipped with mounting hardware, male or female (see table II)

FIGURE 4 Mounting hardware for MIL-C 55302/27

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REQUIREMENTS

Design and construction

Dimensions and configurations See figure 1 and table I

Material.

(E) Insulator body Insulator material shall be in accordance with MIL-C-55302 or MIL-C-24519 type GLCP-30F.

Gasket: Silicon rubber.

Jackscrews and mounting hardware Stainless steel in accordance with MIL-C-55302, passivated in accordance with QQ-P-35.

(E) Contacts: Contact material shall be in accordance with MIL-C-55302 or beryllium copper alloy CDA174 in accordance with ASTM B768

Plating Contacts and their terminations shall be gold plated IAW MIL-C-55302 over nickel plating IAW MIL-C-55302.

Keying See table II

Crimp tool:

AMP hand tool P/N 90346-1, or approved equivalent For use with M55302/113-05 contacts

AMP automatic tool P/N 466625-3, or approved equivalent For use with M55302/113-06 contacts

Contact identification See figure 1

Mating and unmating The maximum mating force, in pounds, shall not exceed a value equal to 0 25 times the number of contacts, and the withdrawal force, in pounds, shall be a minimum of 0 025 times the number of contacts and shall not exceed the measured insertion force, when the housing is loaded with contacts and mated with connector specified in MIL-C-55302/27

Current rating:

3 0 amperes maximum per contact

2 25 amperes continuous per contact at 75°F

Contact resistance No individual contact pair shall have a resistance exceeding 0 020 ohm

Contact retention 3 pounds per contact

Dielectric withstanding voltage

Sea level 900 volts rms

High altitude 200 volts rms.

Part number M55302/113-(dash number from tables I, II, and III)

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TABLE I Housing dash numbers (without hardware), contact positions, and dimensions

Dash number	Number of positions	Contact identification numbers		A	B	C	D	E
-15	30	1 16	15 30	2.355 (59.82)	2.065 (52 45)	14	1 400 (35 56)	N/A
-16	40	1 21	20 40	2 855 (72 52)	2 565 (65 15)	19	1 900 (48 26)	N/A
-17	50	1 26	25 50	3 355 (85 22)	3 065 (77 85)	24	2 400 (60 96)	N/A
-18	60	1 31	30 60	3 855 (97.92)	3 565 (90 55)	29	2 900 (60 96)	N/A
-19	70	1 36	35 70	4 355 (110 62)	4 065 (103.25)	34	3 400 (86 36	N/A
-20	80	1 41	40 80	4 855 (123 32)	4 565 (115 95)	39	3 900 (99 06)	N/A
-21	90	1 46	45 90	5 355 (136.02)	5 065 (128 65)	44	4 400 (111.76)	N/A
-22	100	1 51	50 100	5 855 (148 72)	5 565 (141 35)	49	4 900 (124 46)	N/A
-23	110	1 56	55 110	6 355 (161 42)	6 065 (154 05)	54	5 400 (137 16)	N/A
-88	70	1 36	35 70	4 355 (110 62)	4 065 (103 25)	34	3 400 (86 36)	(E) 3 600 REF (91 44)
-89	110	1 56	55 110	6 355 (161 42)	6 065 (154 05)	54	5 400 (137 16)	(E) 5 600 REF (142 24)

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TABLE II Dash numbers for pin header housing assembly 1/

Assembly dash number	Housing pin number	Keying jackscrews	
		Groove end	Tongue end
-07	-19	-11	-11
-08		-11	-12
-09		-12	-11
-10		-12	-12
-28	-15	-11	-11
-29		-11	-12
-30		-12	-11
-31		-12	-12
-36	-16	-11	-11
-37		-11	-12
-38		-12	-11
-39		-12	-12
-44	-17	-11	-11
-45		-11	-12
-46		-12	-11
-47		-12	-12
-52	-18	-11	-11
-53		-11	-12
-54		-12	-11
-55		-12	-12
-60	-20	-11	-11
-61		-11	-12
-62		-12	-11
-63		-12	-12
-68	-21	-11	-11
-69		-11	-12
-70		-12	-11
-71		-12	-12
-76	-22	-11	-11
-77		-11	-12
-78		-12	-11
-79		-12	-12
-84	-23	-11	-11
-85		-11	-12
-86		-12	-11
-87		-12	-12
-90	-88	-11	-11
-91		-11	-12
-92		-12	-11
-93		-12	-12
-94	-89	-11	-11
-95		-11	-12
-96		-12	-11
-97		-12	-12

1/ Pin header housing assembly consists of housing with jackscrews installed.

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TABLE III Dash number for connector kits and component parts 1/

Kit dash number	Assembly pin header housing (see table II)	Hardware, receptacle mounting (see figure 4)			Contacts (loose piece) 2/
		-14	-13	2-56 nut	-05 3/
-01	-7	2	0	2	70
-02	-8	1	1		
-03	-9	1	1		
-04	-10	0	2		
-24	-28	2	0	2	30
-25	-29	1	1		
-26	-30	1	1		
-27	-31	0	2		
-32	-36	2	0	2	40
-33	-37	1	1		
-34	-38	1	1		
-35	-39	0	2		
-40	-44	2	0	2	50
-41	-45	1	1		
-42	-46	1	1		
-43	-47	0	2		
-48	-52	2	0	2	60
-49	-53	1	1		
-50	-54	1	1		
-51	-55	0	2		
-56	-60	2	0	2	80
-57	-61	1	1		
-58	-62	1	1		
-59	-63	0	2		
-64	-68	2	0	2	90
-65	-69	1	1		
-66	-70	1	1		
-67	-71	0	2		
-72	-76	2	0	2	100
-73	-77	1	1		
-74	-78	1	1		
-75	-79	0	2		
-80	-84	2	0	2	110
-81	-85	1	1		
-82	-86	1	1		
-83	-87	0	2		
-98	-90	2	0	2	70
-99	-91	1	1		
-100	-92	1	1		
-101	-93	0	2		
-102	-94	2	0	2	110
-103	-95	1	1		
-104	-96	1	1		
-105	-97	0	2		

1/ Component parts may be ordered separately.
2/ 06 contacts are supplied in reel form.
3/ Contacts shipped loose in a plastic bag unless otherwise specified.

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CONCLUDING MATERIALS

Custodian
Air Force - 17

Review activities
Air Force - 11, 85, 99
DLA - ES

User activity
Air Force - 19

Preparing activity
Air Force - 17

Agent.
DLA - ES

(Project 5935-F405-01)

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